Lecture # 5,6

Friday, March 6, 2026 9:45 AM

$$\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{cases}$$

z (?)

$\rightarrow$ $e_1x + f_1y = g_1$

$e_2x + f_2y = g_2$

Eliminate x, $y = B$

$x = A$

$z = C$

GAUSS Elimination method

$2x + 3y = 10$

$x - 2y = -2$

WAISAY

$2x + 3y = 10$

$-2x + 4y = 4$

$7y = 14$

$\boxed{y = 2, \; x = 2}$

Augumented Matrix

$$\text{I} \quad \left(\begin{array}{cc|c} 2 & 3 & 10 \\ 1 & -2 & -2 \end{array}\right)$$

As "1"

① R_1, R_2 SWAP $\sim R_{12}$

② $R_1 - R_2$

③ $\frac{1}{2}R_1$

Pivot Element

$$\left(\begin{array}{cc|c} 1 & -2 & -2 \\ 2 & 3 & 10 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & -2 & -2 \\ 0 & 7 & 14 \end{array}\right)$$

$a_1x + b_1y = c_1$

$a_2x + b_2y = c_2$

I Augumented Matrix

$$\left(\begin{array}{cc|c} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{array}\right) \begin{array}{l} \rightarrow R_1 \\ \rightarrow R_2 \end{array}$$

Row operations

$$\text{II} \quad \begin{array}{ccc} x & y & \\ \end{array} \left(\begin{array}{cc|c} 1 & d & e \\ 0 & 1 & f \end{array}\right)$$

III Results

$R_2 \Rightarrow \quad 1y = f \Rightarrow \boxed{y = f}$

$R_1 \quad 1x + dy = e$

$x + df = e$

$\boxed{x = e - df}$

R_2	2	3	10
$2R_1$	2	-4	-4
$\sim R_2 - 2R_1$	0	7	14

$x + y = 5$

* For "1" use same Row or.

$$\left(\begin{array}{cc|c} 0 & 7 & 14 \end{array}\right)$$

$$\begin{array}{cc} x & y \end{array}$$

$$\sim \tfrac{1}{7}R_2 \left(\begin{array}{cc|c} 1 & -2 & -2 \\ 0 & 1 & 2 \end{array}\right)$$

$R_2 \to 1y = 2$ $\quad R_1$ $\quad 1x - 2y = -2$

$\boxed{y = 2}$ $\quad 1x - 4 = -2$

$\boxed{x = 2}$

$x + y = 5$

$x - y = 2$

* For "1" use some Row or Row After This.

$2x + 3y = 5$

$$\begin{array}{cc} -1 & +1 \end{array}$$

$2x + 3y = 10$

$4x + 6y = 15$

$$\left(\begin{array}{cc|c} 2 & 3 & 10 \\ 4 & 6 & 15 \end{array}\right)$$

$$\tfrac{1}{2}R_1 \left(\begin{array}{cc|c} 1 & 3/2 & 5 \\ 4 & 6 & 15 \end{array}\right)$$

$R_2 \quad 4 \quad 6 \quad 15$

$4R_1 \quad \underline{4} \quad \underline{6} \quad \underline{20}$

$0 \quad 0 \quad -5$

$$\sim R_2 - 4R_1 \left(\begin{array}{cc|c} 1 & \frac{3}{2} & 5 \\ 0 & 0 & -5 \end{array}\right)$$

No soln

$0x + 0y \neq -5$

$3x + 4y = 10$

$6x + 8y = 20$

$$\left(\begin{array}{cc|c} 3 & 4 & 10 \\ 6 & 8 & 20 \end{array}\right)$$

$$\tfrac{1}{3}R_1 \left(\begin{array}{cc|c} 1 & \frac{4}{3} & \frac{10}{3} \\ 6 & 8 & 20 \end{array}\right)$$

$R_2 : 6 \quad 8 \quad 20$

$6R_1 : \underline{6} \quad \underline{8} \quad \underline{20}$

$0 \quad 0 \quad 0$

$$R_2 - 6R_1 \left(\begin{array}{cc|c} 1 & \frac{4}{3} & \frac{10}{3} \\ 0 & 0 & 0 \end{array}\right)$$

Infinite Sol

$x + \frac{4}{3}y = \frac{10}{3}$

Infinite Set $x + \frac{4}{3}y = \frac{10}{3}$

$y = t \quad x + \frac{4}{3}t = \frac{10}{3}$

$x = \frac{10}{3} - \frac{4}{3}t$

$\left(\frac{10}{3} - \frac{4}{3}t, t\right)$

Row operations

* You CANNOT Add or Subtract Any particular Number from any Row.

Like $\left[\begin{array}{ccc|c} 2 & 3 & 4 & 5 \end{array}\right]$

$\times \left[\begin{array}{ccc|c} 2\frac{-1}{+} & 3\frac{-1}{+} & 4\frac{-1}{+} & 5\frac{-1}{+} \end{array}\right]$

Can not do

But You Can Add / Subtract Rows with each other

$R_1 + 2R_2, \quad R_2 - 3R_1$

* You CANNOT multiply / divide any Two Rows $\quad R_1 \cdot R_2 \times ; \quad \frac{R_1}{R_2} \times$

But Can multiply any Row by any Constant

$4R_1 \checkmark \longrightarrow$ or $\frac{1}{4}R_1 \checkmark$

* Swaping of Rows $\quad R_{ij} = R_{ji}$

* Alway Create "1" in diagonals and Then create zeros below or above it. Then go for next Row to create 1.

Row Containing '1' is Called Pivot Row

Example 1 Use elementary row operations to solve the linear system:

$x_1 + x_2 + 2x_3 = 9$

$2x_1 + 4x_2 - 3x_3 = 1$

$3x_1 + 6x_2 - 5x_3 = 0$

Augumented Matrix

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array}\right)$$

$$\begin{array}{r} 2 \quad 4 \quad -3 \quad 1 \\ -2 \quad -2 \quad -4 \quad -18 \\ \hline 0 \quad 2 \quad -7 \quad -17 \end{array} \qquad \begin{array}{r} 3 \quad 6 \quad -5 \quad 0 \\ -3 \quad -3 \quad -6 \quad -27 \\ \hline 0 \quad 3 \quad -11 \quad -27 \end{array}$$

$$\begin{array}{l} \\ R_2 - 2R_1 \\ R_3 - 3R_1 \end{array} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{array}\right)$$

$$\begin{array}{r} 0 \quad 3 \quad -11 \quad -27 \\ 0 \quad 3 \quad -\frac{21}{2} \quad -\frac{51}{2} \\ \hline \end{array}$$

$-11 + \frac{21}{2}$

$-27 + \frac{51}{2}$

$$\begin{array}{l} \\ \frac{1}{2}R_2 \\ \\ \end{array} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 3 & -11 & -27 \end{array}\right)$$

$$\begin{array}{l} \\ \\ R_3 - 3R_2 \end{array} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & -\frac{1}{2} & -\frac{3}{2} \end{array}\right)$$

$$\begin{array}{l} \\ \\ -2R_3 \end{array} \begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} \\ \left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{array}\right) \end{array}$$

$0x_1 + 0x_2 + 1x_3 = 3$

$\boxed{x_3 = 3}$

R_2: $\quad x_2 - \frac{7}{2}x_3 = -\frac{17}{2}$

$x_1 \quad x_2 \quad x_3$

$x_2 \cdot$

$x_2 \quad \frac{1}{2}x_3 - \frac{1}{2}$

$\begin{matrix} x_1 & x_2 & x_3 \\ \{1, & 2, & 3\} \end{matrix}$

$x_2 - \frac{21}{2} = -\frac{17}{2}$

$x_2 = -\frac{17}{2} + \frac{21}{2} = 2$

$x_1 + x_2 + 2x_3 = 9$

$x_1 + 2 + 6 = 9$

$x_1 = 9 - 8$

$x_1 = 1$

$$\begin{cases} x + z + 2w = 6 \longrightarrow (1) \\ y - 2z = -3 \longrightarrow (2) \\ x + 2y - z = -2 \longrightarrow (3) \\ 2x + y + 3z - 2w = 0 \longrightarrow (4) \end{cases}$$

$$\begin{array}{c} \begin{array}{cccc} x & y & z & w \end{array} \\ \left(\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 6 \\ 0 & 1 & -2 & 0 & -3 \\ 1 & 2 & -1 & 0 & -2 \\ 2 & 1 & 3 & -2 & 0 \end{array}\right) \end{array}$$